



Document	User Manual
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Document Control

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Revision History

Revision	Date	Prepared By	Summary
1.0.0	May 9, 2026	Paco Salas DRH	Initial final end-user manual for product version 1.0.0.

Copyright and Disclaimer

Important: Support details, legal notices, and distribution terms should remain aligned with the released package version.

Topic	Statement
License	GPL-3.0-or-later.
Copyright Holder	Paco Salas DRH.
Warranty	Unless separately stated in writing, the software is provided without warranty.
Commercial Distribution	The package notice allows commercial distribution, provided GPL terms, source availability, and included notices remain intact.
Manual Legal Status	This manual is intended for end-user guidance and product distribution documentation.

First-Run Quick Start

Use this sequence to reach a reliable first result with confidence:

1. Install and enable the add-on in Blender.
2. Open the Node Editor sidebar and select the DRH-Ramp Studio tab.
3. Make a native Color Ramp node active and generate a preset ramp.
4. Confirm the preview updates, then test Convert Selection or Color Sampler.

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About This Manual

This manual provides end-user guidance for DRH - Color Ramp Studio, a Blender add-on for generating, converting, sampling, refining, previewing, and transferring Color Ramp nodes in supported node editors.

Scope

This manual covers installation, configuration, everyday operation, supported workflows, temporary data behavior, known limits, troubleshooting, and support guidance for DRH - Color Ramp Studio version 1.0.0.

Conventions

Conventions Used	Meaning
Bold UI labels	UI labels such as Generate Ramp, Convert Selection, Color Sampler, and Capture Backend Check.
Numbered steps	Procedures that should be followed in order for a reliable result.
Note / Tip / Warning / Important	Callouts that highlight practical guidance, workflow cautions, or platform-specific behavior.
Important	A behavior or limitation that should be reviewed before continuing.
Figure	Reserved for interface illustrations when screenshots are included in future editions.

Note: UI labels and workflow names in this manual follow the released package and should be treated as the operational reference for version 1.0.0.

Audience and Intended Use

This manual is intended for Blender users working with Color Ramp nodes across shader, compositor, and geometry-node workflows. It is especially suited to artists, technical artists, reviewers, and production users who need dependable generation, editing, transfer, sampling, cleanup, and restoration tools.

Product Overview

DRH - Color Ramp Studio is a Blender add-on focused on practical Color Ramp workflows. It appears in Node Editor > Sidebar > DRH-Ramp Studio and provides generation, conversion, sampling, transfer, and cleanup tools around native Blender Color Ramp nodes.

Its main user journeys are creating ramps from presets or palette logic, extracting colors from images or the screen, refining stop placement and color flow, and moving between native ramps and group-based editing workflows when more control is needed.

Who It Is For

Material, shading, and look-development artists who build reusable gradients and masks.

Geometry Nodes users who need Color Ramp generation, cleanup, or conversion tools inside node-based workflows.

Compositing users who need faster palette construction, transfer, and screen-sampled gradients.

1. Technical artists and support reviewers who need predictable ramp restoration behavior and documented platform limitations.

Important: Many operators require an active supported node. Generation and preview require an active native Color Ramp node, while several editing tools also support converted groups.

Capabilities

Area	What It Does
Ramp generation	Builds native Color Ramp nodes from presets, random palettes, grayscale palettes, stripe patterns, complementary or analogous pairs, and sampled image palettes.
Preset browsing	Provides a curated preset library with searchable preview thumbnails and dominant-color filtering.
Conversion and restore	Converts native ramps to editable group-based ramps and restores them back to native Color Ramp nodes when supported.
Spline-aware handling	Warns when Cardinal or B-Spline ramps require baked sampling and offers restore or baked-linear return choices.
Transfer tools	Copies ramp data to an internal studio clipboard and pastes it to another supported ramp target.
Stop refinement	Redistributes, normalizes, randomizes, rounds, simplifies, duplicates, or de-duplicates ramp stops.
Color design tools	Applies luminance sorting, hue sorting, inversion, mirroring, harmony expansion, value shaping, and palette mutation helpers.
Cleanup utilities	Redistributes, normalizes, simplifies, randomizes, reverses, de-duplicates, and refines stops.
Preview helpers	Supports ramp review, helper actions, and day-to-day editing, Color Flow, and design-assist tasks.
Screen and image sampling	Samples gradients from the screen, loads reference images, and extracts representative colors from image files.

Key Features

Feature	Details
100 curated preset palettes	The released package includes 100 preset palettes with thumbnail previews.
Color Sampler	Capture on-screen gradients with overlay preview, subject to operating-system support.
Configurable panel location	The N-panel category can be renamed from add-on preferences.
Context-menu helpers	Access selected tools directly from supported node context menus and assistive workflow menus.

Feature	Details
Editable group-based conversion	Advanced editing path for workflows that need more control than a native ramp alone.
Helper node generation	Converted groups can create helper color nodes appropriate to the current node-tree context.
Local-only normal operation	Normal operation does not require internet access or an external service.
Native Color Ramp generation tools	Fast creation of ramps inside supported Blender node editors.
Reference image helpers	Loads a temporary reference image into Blender and can open or close a dedicated Image Editor area.
Restore tools	Return converted ramps to native ramps when appropriate.
Screen-capture backend check	The sampler tools include a backend check that validates the bundled capture backend and attempts a tiny real capture.
Tabbed or stacked interface	The main panel can be used in tab mode or expanded stacked sections.
Transfer and utility tools	Copy, paste, reverse, normalize, randomize, simplify, and refine ramp data.

System Requirements

Requirement	Verified Information
Host application	Blender 4.2.0 or later
5.x note	Blender 5.x behavior may depend on add-on metadata and Blender API changes.
Supported platforms	Windows x64, macOS x64, macOS arm64, and Linux x64 are declared in the add-on manifest.
Dependencies	No external user-installed dependency is documented for normal use.
Bundled binaries	No external binary bundle is documented. The package includes vendored Python source for the mss screen-capture dependency.
Filesystem access	Reads user-selected image files and writes temporary preset preview thumbnails to the system temporary directory when the preset browser is opened.
Internet access	Not required for normal operation.
Disk space / memory	No separate package requirement is documented beyond normal Blender scene and image-processing needs.

Compatibility

Area	Compatibility Notes
Host application	Blender 4.2.0 or later is declared by the manifest and bl_info metadata.

Area	Compatibility Notes
Windows	Supported by the packaged manifest.
macOS	Supported by the packaged manifest. Screen sampling may require Screen Recording permission for Blender.
Linux	Supported by the packaged manifest. The Color Sampler is not supported in pure Wayland sessions with the bundled backend; X11 or XWayland is required.
File format or data compatibility	Works with supported Color Ramp workflows in Shader Editor, Geometry Nodes, and Compositor. Image-based generation accepts common readable image formats documented by the package.
Installation or discovery logic	Install from the packaged ZIP through Blender's extension or add-on installer, depending on Blender version and setup.

Supported Scope and Non-Supported Scope

Warning: The Color Sampler depends on OS-level capture support. On Linux, the bundled backend does not support pure Wayland sessions.

DRH - Color Ramp Studio is designed to work on Color Ramp generation, editing, transfer, and conversion inside supported Blender node editors. It is not a general-purpose color-management system, it does not rely on cloud services, and it should not be assumed to guarantee lossless round-trips for sampled spline conversions or identical behavior across every future Blender release.

Core workflows operate within supported Blender node editors and supported Color Ramp targets.

Group-based ramps prioritize flexibility, deeper editing control, and recoverability over exact native parity.

Screen sampling is platform-dependent and requires the appropriate operating-system permissions and supported session backends.

On Linux, the sampler does not support pure Wayland sessions; X11 or XWayland is required for screen-capture functionality.

Included Package Components

Component	Purpose
__init__.py and registration.py	Package metadata and Blender registration logic.
panels.py	Main Node Editor sidebar panel and section layout.
operators.py	Generation, conversion, transfer, refinement, and design operators.
properties.py	Scene and runtime properties used by the interface and operators.
preferences.py	Add-on preferences, defaults, panel category settings, and Linux capture backend option.
sampler.py	Color Sampler tools, reference-image helpers, context-menu integration, and backend check.

Component	Purpose
capture_utils.py and stroke_utils.py	Screen-capture helpers and stroke-processing utilities.
core.py and ramp_utils.py	Shared ramp logic, palette data, interpolation handling, and utility functions.
mss/	Vendored python-mss dependency used by the Color Sampler feature.
Readme.md, support.md, changelog.md	User-facing package guidance, support notes, and release summary.
License.txt, notice.txt, copyright.txt, third_party_notices.txt	License, notice, attribution, and redistribution documentation.

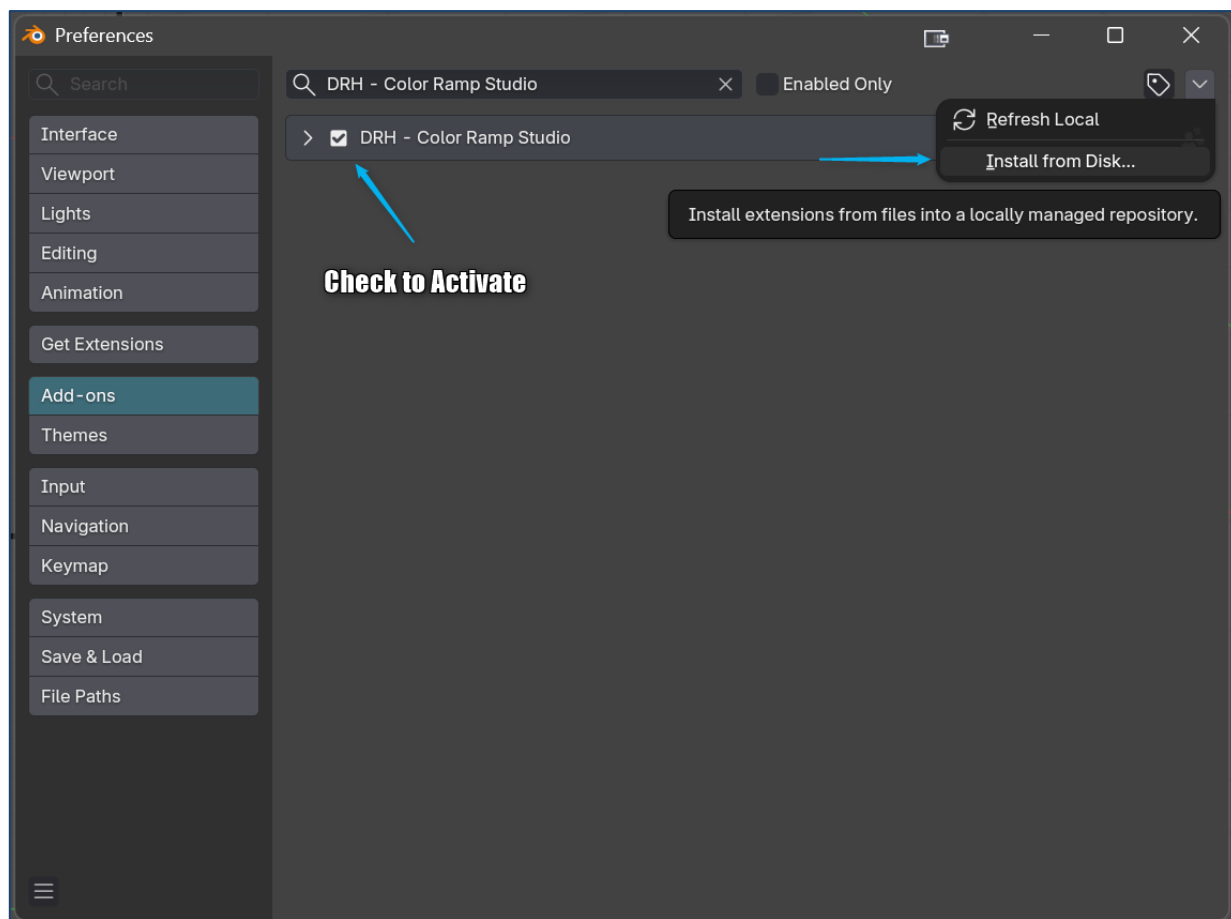
Installation and Setup

Prerequisites

1. Use Blender 4.2.0 or later in a supported environment.
2. Keep the packaged ZIP intact for installation from disk or extensions workflow.
3. Confirm you can enable add-ons or extensions in the current Blender installation.
4. If you plan to use Color Sampler, confirm operating-system screen-capture permissions before first use.

Standard Installation Method

1. Open Blender.
2. Open Edit > Preferences.
3. Use Get Extensions or Install from Disk / Install Add-on from Disk, depending on your Blender version and setup.
4. Select the DRH - Color Ramp Studio ZIP package.
5. Enable DRH - Color Ramp Studio.
6. Open the Node Editor sidebar.
7. Confirm that the DRH-Ramp Studio tab and panel are available.



Note: If the panel does not appear immediately, confirm the add-on is enabled and open the Node Editor sidebar with N.

Alternative Installation or Extension Manager

1. In Blender builds that use the Extensions workflow, install the packaged ZIP through the Extensions interface and then enable the add-on.
2. In Blender builds that use the classic add-on installer, use Install from Disk / Install Add-on from Disk and then enable the package.
3. Verify the entry point at Node Editor > Sidebar > DRH-Ramp Studio.

Important: Blender menu wording differs by version and deployment style, but the supported installation source is the packaged ZIP.

First Launch and Verification

1. Open a supported Node Editor workspace.
2. Make a native Color Ramp node active or run Create Color Sampler to create one.
3. Generate a preset ramp or open the preview to confirm the add-on is responding to the active node.
4. Review add-on preferences if you want to change tab mode, context-menu visibility, panel category, or Linux capture backend.

Update, Reinstall, and Uninstall

Task	Recommended Procedure
Update	Install the newer packaged ZIP through the same Blender workflow, then confirm the panel, preferences, and sampler backend still behave as expected.
Reinstall	Disable the add-on, reinstall the packaged ZIP, re-enable it, and verify the panel entry and active operators.
Uninstall	Disable the add-on in Blender and remove it through the Extensions or Add-ons preferences interface used for installation.
Preserve outputs	Scene content such as native ramps or converted groups remains in the .blend file. Temporary preset preview thumbnails and temporary reference-image session data are not part of a long-term export workflow.

Common Installation Problems

Issue	Likely Cause	Resolution
The panel does not appear in the Node Editor sidebar.	The add-on is not enabled or the user is not in a supported editor.	Enable the add-on in Preferences and open the Node Editor sidebar in a supported node editor.
Install from Disk fails or the package is not recognized.	The ZIP was altered, unpacked, or selected from the wrong location.	Re-download or reuse the intact packaged ZIP and install that file directly.
The sidebar tab is missing	Wrong editor context or panel not enabled	Open the Node Editor sidebar and confirm the add-on is enabled and the panel is not hidden.
The Color Sampler cannot capture the screen.	OS permissions or Linux session support prevent screen capture.	Run Capture Backend Check, verify permissions, and on Linux use X11 or XWayland rather than a pure Wayland session.
The image-based generator produces no useful result.	The image path is invalid or the source cannot be read into Blender.	Choose a readable image file and test with a different image when the source is fully transparent or unsuitable.

Operational Details

Area	Product Guidance
Activation	No activation or entitlement workflow is documented for this package.
License model	The package is distributed under GPL-3.0-or-later.
Commercial support terms	Separate paid services may be offered, but software distribution itself is provided without warranty unless separately stated in writing.
Initial setup tasks	Enable the add-on, confirm the DRH-Ramp Studio panel appears, and review preferences if you want to change layout defaults or sampler backend behavior.

The shortest reliable first result is to make a native Color Ramp node active, open the Generator section, choose a preset, and run Generate Ramp.

Quick Start

This sequence takes a first-time user from installation to a confirmed working ramp inside the Node Editor.

Step	Action	What to Confirm
1	Install and enable DRH - Color Ramp Studio.	The add-on is enabled in Blender Preferences.
2	Open the Node Editor sidebar and select DRH-Ramp Studio.	The Generator, Conversion, or Tools interface is visible.
3	Make a native Color Ramp node active.	The Preview section shows the active ramp and generator controls become available.
4	Choose a preset and run Generate Ramp.	The active Color Ramp changes to the selected preset or generated palette.
5	Optionally run Convert Selection or Capture Backend Check.	You can confirm conversion or sampler-related workflows before deeper production use.

Add onboarding notes only when they provide practical first-session guidance that improves success for new users. Start with a preset ramp before testing conversion, image extraction, or screen sampling.

After enabling the add-on, open a supported Node Editor, select or add a Color Ramp node, open the DRH-Ramp Studio tab in the sidebar, and run a simple generator or utility action. When you load a reference image, the popup offers OK to close or Open Image Editor Window for a dedicated preview. Launch Color Sampler from the Nodes N-panel when you are ready to capture an on-screen gradient.

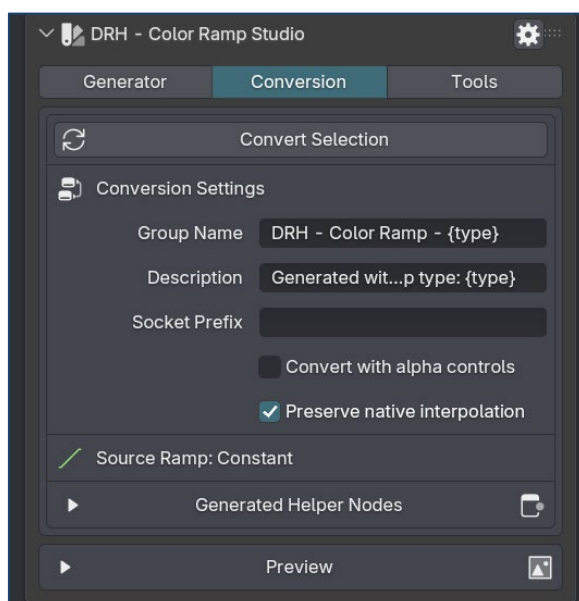
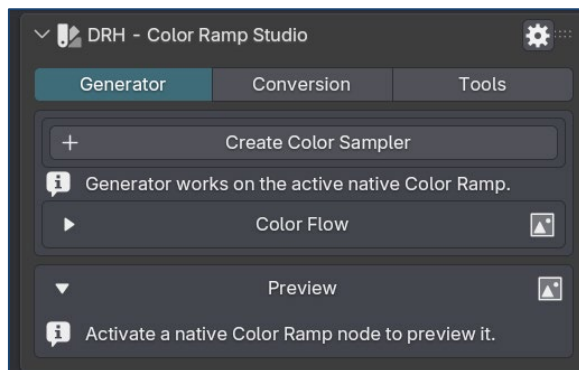
Product Preferences and Settings

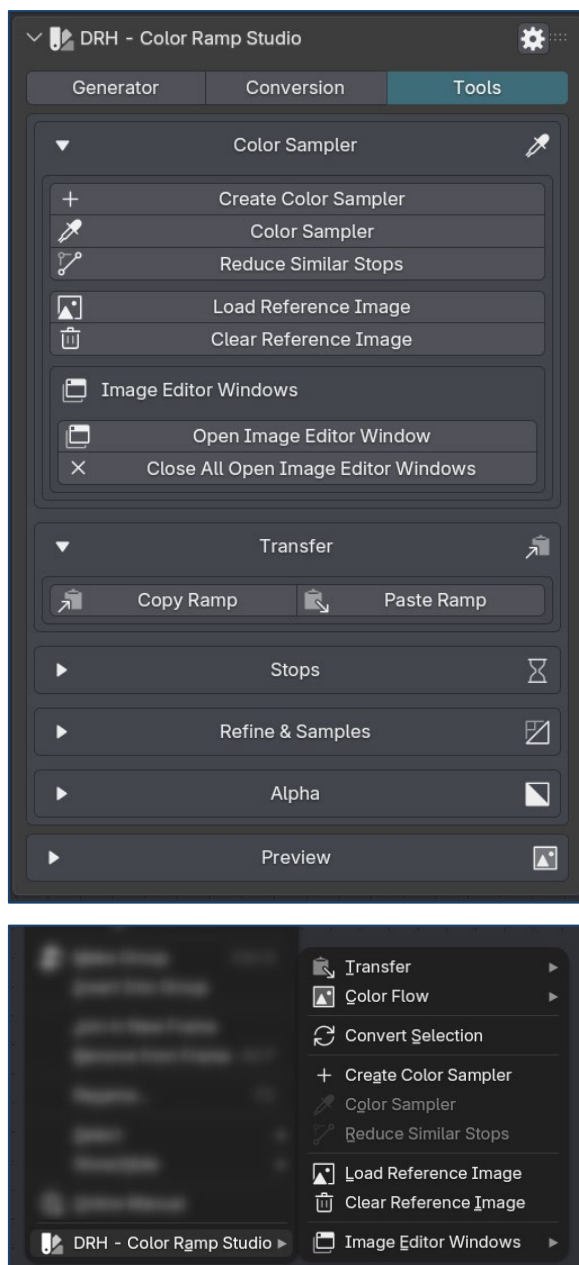
DRH - Color Ramp Studio includes both Blender-level add-on preferences and scene-level runtime settings that control panel layout, generation behavior, conversion defaults, and capture-related options.

Host Application Preferences / Extension Manager

Preferences Section	Purpose
Add-on entry in Blender Preferences	Enables the package and exposes preferences for panel defaults, generator defaults, conversion defaults, and tools.
N Panel Category	Renames the sidebar tab used by the add-on.
Use tabs by default	Shows Generator, Conversion, and Tools as tabs instead of stacked sections.
Generator	Stores preferred generator-related defaults.
Conversion	Controls conversion behavior and restore preferences.
Tools	Holds defaults for transfer, sampler, and stop-editing helpers.
Enable Context Menu	Adds the DRH - Color Ramp Studio tools to supported node context menus.

Preferences Section	Purpose
Panel defaults	Controls which major sections open by default in stacked mode.
Linux Capture Backend	On Linux, selects the preferred bundled capture backend for the Color Sampler tools.





Runtime Panel Settings

Setting	Description
Palette Source	Chooses whether generation is based on a preset, image, random palette, grayscale palette, stripes, or paired-color logic such as complementary or analogous.
Preset / preset browser	Selects a curated palette and optionally shows searchable preview thumbnails.
Show as Tabs	Switches the main panel between tabbed and expanded section presentation.
Enable Context Menu	Adds supported helper actions to node context menus.
Generator section state	Controls whether generator tools are open by default.
Conversion section state	Controls whether conversion tools are open by default.

Setting	Description
Tools section state	Controls whether utility tools are open by default.
Interpolation	Sets the interpolation used by a generated native Color Ramp.
Distribution	Controls spacing behavior for generated or redistributed stop positions.
Stops	Sets the target stop count for generation or simplification workflows.
Random Positions	Randomizes generated stop positions while keeping endpoints stable.
Group Name / Description / Socket Prefix	Controls metadata and socket naming for converted group-based ramps.
Convert with alpha controls	Exposes dedicated alpha sockets and alpha output during conversion when enabled.
Stop decimals / generated decimals	Limits stop-position precision for generated or edited ramps.

Default Rules and Status Logic

The add-on surfaces a small set of practical runtime states and warnings that help users choose the correct workflow before committing a change.

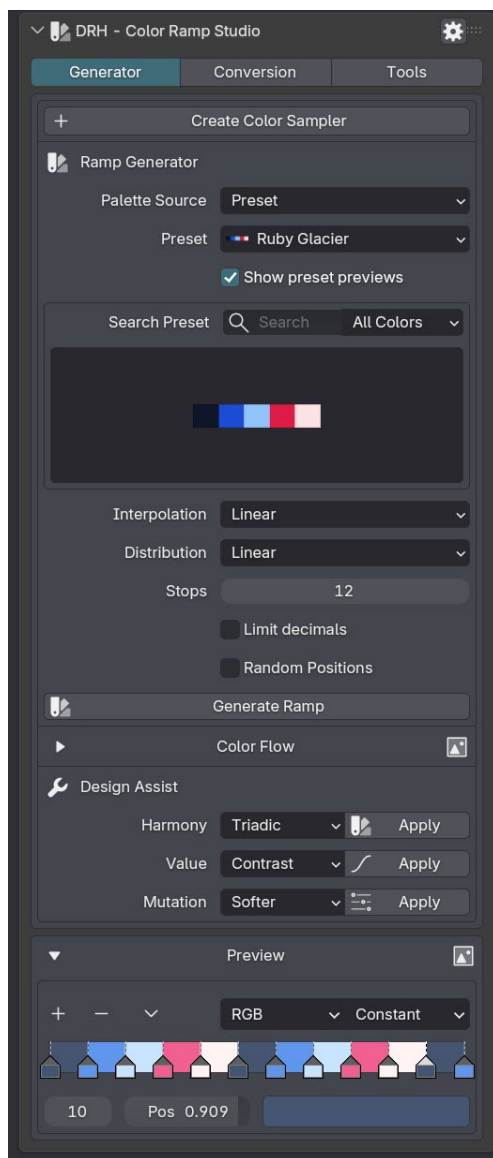
State or Status	Meaning or Default Behavior
Active native Color Ramp	Generator and preview tools work directly on the active native Color Ramp node.
Converted group detected	Conversion tools switch to a restore-oriented workflow and the panel suggests Convert Back to Color Ramp.
Cardinal or B-Spline source ramp	The conversion dialog warns that a baked sample fallback may be used for round-tripping.
Ease interpolation source ramp	Conversion maps Ease behavior to Smooth Step group blending.
Readable image path	Image-based palette extraction requires a readable image file path.
Capture permission available	Color Sampler requires operating-system permission and supported backend conditions.

Important: When converting Cardinal or B-Spline ramps, review the dialog choice carefully. Restore Original Spline returns the stored native spline, while Keep Baked Edits as Linear returns the edited baked result as a native linear ramp.

Interface Overview

In Blender, the add-on appears in Node Editor > Sidebar > DRH-Ramp Studio. The interface can be shown as tabs or stacked sections and centers around Generator, Conversion, Tools, and Preview behaviors.

Area	What It Contains
Generator	Palette-source selection, preset browser, generation options, color-flow tools, and design-assist operators.
Conversion	Convert Selection or Convert Back actions plus naming, interpolation, alpha, and helper-node options.
Tools	Color Sampler, transfer tools, stop tools, simplify tools, alpha tools, and reference-image helpers.
Preview	A live preview of the active native Color Ramp when one is selected.



Visual Interface Walkthrough

The interface is designed for day-to-day ramp work: generate or adjust a native ramp, convert it when advanced group editing is needed, then use the tool sections for transfer, cleanup, sampling, and refinement. When screenshots are not included, the tables in this section describe the practical control layout and entry points.

Main Panels, Menus, and Controls

Primary Actions

Control	Function
Generate Ramp	Applies the selected generator source to the active native Color Ramp.
Convert Selection / Convert Back to Color Ramp	Converts selected native ramps to editable groups or restores supported converted groups back to native ramps.
Create Color Sampler	Adds a new Color Ramp node, makes it active, and opens the Color Sampler workflow.
Color Sampler	Draws a screen stroke to sample colors into the active Color Ramp.
Copy Ramp / Paste Ramp	Copies ramp data into the studio clipboard and pastes it onto another supported target.
Reset All	Restores supported settings or defaults for a clean working state.

Primary Setup

Field	Purpose
Palette Source	Selects the generation method or working mode used for ramp creation.
Preset	Chooses a curated preset when the Preset source is active.
From / To	Defines the anchor pair used by Complementary or Analogous generation.
Image	Selects the source image used for image-based palette extraction.
Group Name / Description	Controls naming and descriptive metadata for converted groups.
Socket Prefix	Adds an optional prefix to generated group sockets.
Default sample count	Sets the typical stop count or sampling granularity for relevant tools.

Preset Browser and Filter Controls

Control	Purpose
Show preset previews	Displays thumbnail previews for curated presets.
Search Preset	Filters the preset list by name or color-tag text.
Color filter	Limits preset results to a dominant color family such as blue, green, neutral, or warm groups.
Thumbnail grid	Provides a visual selection grid for preset palettes.
Preset selector	Sets the current preset used by Generate Ramp.
No preset matches message	Warns when the current search and color filters exclude every preset.

Active Ramp Preview and Current Context

Displayed Value	Notes
Preview strip	Shows the active native Color Ramp directly inside the panel when a native ramp is selected.
Source Ramp	Displays the source interpolation type during conversion workflows.
Bake preview	Shows the planned sampled-stop count for Cardinal or B-Spline conversion workflows.
Selection context	Indicates whether the current target is a native Color Ramp, converted group, or unsupported selection.
Generator target	Generation operates on the active native Color Ramp only.
Tool target	Many tools operate on either a native ramp or a converted group, depending on operator support.

Reference Image and Editor Controls

Control	Function
Load Reference Image	Loads a temporary reference image into Blender for side-by-side visual guidance.
Clear Reference Image	Removes the temporary reference image from Blender.
Open Image Editor Area	Opens a new Image Editor area in the current screen for sampler reference work.
Close Sampler Image Editor	Closes the Image Editor area opened by the sampler tools in the current screen.
Capture Backend Check	Validates the bundled screen-capture backend and attempts a tiny real capture.
Context menu entry	Provides access to supported DRH - Color Ramp Studio tools from node context menus when enabled in preferences.

Recommended Workflows

Workflow 1: Generate a Ramp

1. Open a supported node editor and make a native Color Ramp node active.
2. Open DRH - Color Ramp Studio and choose the Generator tab or section.
3. Select a source such as Preset, Complementary, Analogous, Random, Greyscale, Stripes, or Image.
4. Adjust interpolation, distribution, stop count, and decimal behavior as needed.
5. Run Generate Ramp and review the preview on the active ramp.

Workflow 2: Convert and Restore a Ramp

1. Select a supported native Color Ramp node.
2. Open the Conversion tab or section and review group naming and alpha options.

3. Run Convert Selection.
4. If a warning dialog appears for sampled splines, choose the appropriate return mode before continuing.
5. Edit the converted group as needed for the chosen workflow.
6. Use Convert Back to Color Ramp when you want to return to a native ramp workflow.

Tip: Enable Rename target from preset when building reusable ramp libraries, and keep a clear Group Name template when converted groups will be reviewed by other users.

Workflow 3: Sample a Gradient from the Screen

1. Open the Tools tab or section and confirm a supported ramp target is active.
2. Run Capture Backend Check if you have not yet verified capture support on the current system.
3. Start Color Sampler and draw the screen stroke over the gradient you want to capture.
4. Review the resulting stops and use cleanup tools such as Merge Similar Neighboring Stops or Simplify Samples if needed.

Important: Merge Similar Neighboring Stops and simplify operations immediately change the active ramp. Review subtle gradients before applying these helpers in production work.

Workflow 4: Extract a Palette from an Image

1. Make a native Color Ramp node active.
2. Set Palette Source to Image.
3. Choose a readable image file.
4. Review generation options such as interpolation, distribution, stop count, and decimal behavior.
5. Run Generate Ramp to extract representative colors from the full image.
6. Review the result and adjust with stop or color-flow tools if needed.

Workflow 5: Transfer, Cleanup, and Restore

1. Copy a ramp you trust, then paste it into the target that needs to match it.
2. Use cleanup tools such as normalize, simplify, reverse, stop redistribution, or Color Flow refinement when needed.
3. Restore converted groups to native ramps when a simpler final delivery is preferred.

These workflows cover the main supported day-to-day use of the add-on.

Step-by-Step Usage Instructions

1. Start or Refresh the Main View

Open a supported node editor, select the DRH-Ramp Studio panel, and make sure the intended Color Ramp or converted group is active before running an operator.

Note: Most operators act on the active node selection. If a tool appears inactive or unavailable, first confirm the correct Color Ramp or converted group is selected.

2. Interpret Generation and Conversion Feedback

Feedback Item	Meaning
Preset preview grid	Shows curated preset options when preset previews are enabled.
No preset matches the current filters	The search text or color filter excludes all preset results.
Source Ramp	Reports the interpolation type of the active native ramp before conversion.
Bake preview	Estimates the sampled-stop result for sampled spline conversion workflows.
Converted group detected	Indicates that the active selection is already in the converted workflow state.
Capture Backend Check	Confirms whether the sampler backend initialized and completed a tiny probe capture.
Reference Image Loaded	Confirms that a temporary reference image has been added to Blender.
Preview strip	Lets you confirm whether the active native ramp changed after generation or editing.
Dialog warnings	Explain potential lossiness or workflow implications before a conversion continues.
Inactive operator state	Usually indicates that the current selection or editor context does not meet the operator requirements.
Stop count target	The active generator or cleanup tool has a defined stop count or simplification goal.
Sampler permission granted	The operating system is allowing the required capture operation.

3. Review Working States

State	Operational Meaning
Native Color Ramp active	Generator, preview, and most editing tools can work directly on the selected ramp.
Converted group active	Conversion actions switch to restore-oriented behavior, and some editing tools continue to work on the converted result.
No supported node selected	Generation and preview are unavailable until a valid node is active.
Helper nodes enabled	Conversion may generate context-appropriate helper nodes alongside the converted group.
Random Positions enabled	Generated positions can vary from run to run while endpoints remain stable.
Limit decimals enabled	Generated or edited stop positions are rounded to the selected precision.

State	Operational Meaning
Reference image loaded	A temporary reference image is available inside Blender for visual guidance.
Image missing	The referenced image path is invalid or unreadable.
Sampler backend verified	The platform-specific capture backend passed the probe capture test.
Error reported	The operator detected a problem and returned an explicit error rather than generating unrelated output.

4. Open Reference Assets and Verification Tools

1. Use Load Reference Image when you want a temporary in-Blender image reference for palette or gradient work.
2. Use Open Image Editor Area if you want a dedicated editor for the loaded reference image.
3. Use Capture Backend Check before screen sampling on a new workstation or after changing OS permissions.

Warning: Preset preview thumbnails are written to the system temporary directory, and screen capture depends on operating-system permissions and session support.

Inputs, Outputs, and Workflow Modes

Working Data and Temporary Assets

Output or Working Data	Purpose or Use Case
Generated native Color Ramp	Primary result of the Generator workflow.
Converted group-based ramp	Editable group representation used for advanced ramp workflows and restoration support.
Stored original ramp snapshot	Internal data used to restore supported converted ramps back to native behavior.
Studio clipboard data	Scene-level temporary clipboard used by Copy Ramp and Paste Ramp.
Preset preview thumbnails	Temporary thumbnail images written to the system temp directory when the preset browser is opened.
Temporary reference image	In-Blender image data used for side-by-side review during sampler or palette workflows.
Helper nodes	Optional nodes created to expose color or position control more conveniently in converted workflows.

Supported Input Sources or Baseline Inputs

Image-based generation and reference-image workflows use readable image files selected by the user. The add-on does not document a general import pipeline beyond these supported image sources.

Accepted File Type	Notes
PNG	Supported for image-based generation and reference-image workflows when Blender can read the file.
JPEG / JPG	Supported for image-based generation when the file is readable.
JP2	Supported when readable by Blender in the current environment.
TIFF / TIF	Supported when readable by Blender in the current environment.
BMP / TGA	Supported when readable by Blender in the current environment.
EXR / HDR / WEBP / GIF	Accepted by the documented file filter for image-based workflows.

Generator Sources and Working Modes

Mode or Profile	Description	Typical Use
Preset	Uses one of the curated preset palettes included with the package.	Fast look development and repeatable palette selection.
Complementary	Builds a palette from paired complementary anchor colors.	High-contrast gradients and quick opposing-color studies.
Analogous	Builds a palette from neighboring hues around the chosen anchor.	Smooth color families and related-gradient design.
Greyscale	Generates grayscale-focused color stops.	Masks, value ramps, and non-color utility gradients.
Image	Extracts representative colors from a readable image file.	Palette derivation from concept art, references, or textures.
Random	Generates random colors.	Idea generation and quick experimentation.
Stripes	Generates black-and-white stripe patterns.	Graphic masks and high-contrast utility ramps.
Conversion	Converts native Color Ramp nodes into editable group-based ramps.	Advanced editing workflows and spline-aware restoration.
Color Sampler	Samples colors from the screen into the active ramp.	Rebuilding gradients from references or UI captures.

Preset Browser Filters

Preset Browser Filter	Included Fields or Behavior
Search Preset	Filters the preset browser by preset name or color-tag text.
Color family filter	Limits results to families such as red, green, blue, brown, gray, black, or white.
Show preset previews	Displays or hides thumbnail previews for curated presets.

Understanding Outputs, Statuses, and Decision Logic

Outputs in DRH - Color Ramp Studio are primarily working node data inside the current Blender session or .blend file. The most important decision points are whether to stay in a native ramp workflow, move to a converted group workflow, or use sampling and cleanup helpers to reach the desired gradient.

Decision Guidance

1. Use Preset or Complementary/Analogous generation when you need a controlled starting palette quickly.
2. Use Image generation when the palette should be derived from a reference image or texture.
3. Use Convert Selection when native Color Ramp editing is not sufficient for the workflow you need.
4. Use Convert Back to Color Ramp when you want to return to a native node and the stored snapshot or baked-linear return mode matches your intent.
5. Use Color Sampler only after capture support has been verified on the current platform.

Output Naming Behavior

Generated native ramps may be renamed to "Preset: <Preset Name>" when Rename target from preset is enabled. Converted groups use the Group Name template, which supports values such as "{type}" and "{node}". Temporary preset preview thumbnails are internal working files in the system temp directory rather than user-facing deliverables.

Note: Keep Group Name and Socket Prefix settings consistent when converted groups will be shared, reviewed, or reused across multiple files.

Privacy, Data Handling, and Local Processing

Normal operation is local. The add-on reads user-selected image files for palette extraction and reference-image workflows, writes temporary preset preview thumbnails to the system temporary directory when previews are opened, stores temporary working data such as the studio clipboard and conversion snapshots inside Blender data structures, and does not require internet access or an external service. Users should review screenshots, shared .blend files, and temporary reference images before sharing them externally if they contain project-sensitive visual material.

Common Use Cases

Use Case	How the Product Helps
Build a reusable shading gradient	Preset generation and distribution controls provide a fast starting point for material look development.
Refine a sampled or noisy ramp	Simplify Samples, Merge Similar Neighboring Stops, and decimal-limiting tools help clean up ramp data.
Derive a palette from concept art	Image-based generation extracts representative colors from a readable reference image.
Recreate a gradient seen on screen	Color Sampler captures colors into the active ramp after backend support is verified.

Use Case	How the Product Helps
Move between native and advanced editing workflows	Conversion and restore tools support a controlled handoff between native ramps and group-based editing.

Each use case maps to a concrete operator path in the main panel so first-time users can reach a result without leaving the Node Editor.

Advanced Tips, Best Practices, and Recommendations

Work with a native Color Ramp first, then convert only when the native node no longer fits the editing task. Keep preset previews enabled when exploring the bundled palette library, but disable them on slower systems if you want a lighter browser experience. Before large cleanup actions, duplicate the ramp or save the .blend file so you can compare subtle gradient changes. On macOS and Linux workstations, decide the preferred capture backend in add-on preferences before relying on the sampler in production.

Update and Migration Guidance

After updating or reinstalling the add-on, verify the panel location, context-menu visibility, capture backend behavior, and any preferred defaults such as tab mode, generator defaults, spline return mode, and helper-node settings. When reopening existing files, confirm whether converted groups still match the intended workflow and review any version-sensitive compositor helper behavior in newer Blender releases.

Troubleshooting

Symptom or Message	Probable Cause	Recommended Action
The Generator controls do not appear usable.	No native Color Ramp node is active.	Select or create a native Color Ramp node, then reopen the Generator section.
Convert Selection is unavailable.	The selection is not a supported Color Ramp context.	Move to a supported node editor and select a valid native Color Ramp or converted group.
The Color Sampler cannot capture the screen.	Capture permissions, session support, or backend initialization failed.	Run Capture Backend Check, grant OS permissions, and on Linux use X11 or XWayland instead of a pure Wayland session.
The image-based generator produces weak or empty results.	The image path is invalid, unreadable, or visually unsuitable.	Verify the file path, test with a different readable image, and avoid fully transparent sources.
A converted ramp does not match the original spline exactly.	The workflow required baked sampling for Cardinal or B-Spline behavior.	Review the conversion dialog and choose the correct restore or baked-linear return mode for the intended result.
No preset matches the current filters.	Search text or color filtering is too restrictive.	Clear the search text or switch the color filter back to All Colors.

Symptom or Message	Probable Cause	Recommended Action
Reference image workflow feels incomplete.	Loading a reference image does not automatically split the workspace.	Use Open Image Editor Area to create a dedicated preview area.
Cleanup changed a subtle gradient too much.	The merge or simplify tools are heuristic helpers.	Undo the action, reduce the cleanup intensity, or review the ramp manually before reapplying.
The panel appears under an unexpected tab name.	The N Panel Category preference was changed.	Open add-on preferences and reset or rename the panel category.

Known Constraints and Operational Notes

1. Results depend on the active Blender session, selected node, and current node-tree context. Save your .blend file whenever you want a stable baseline for comparison or rollback.
2. Platform-sensitive features, especially screen sampling, may behave differently when required permissions or supported display backends are not available.
3. Pure Wayland sessions on Linux are not supported by the bundled Color Sampler backend.
4. macOS may block screen sampling until Screen Recording permission is granted to Blender.
5. Image-based generation requires a valid readable image path.
6. When requested stop-position precision cannot represent all selected stops uniquely, the add-on preserves the original spacing instead of corrupting the ramp.
7. Converting Cardinal or B-Spline ramps may involve a baked sample fallback and should be reviewed before committing the workflow.
8. Merge Similar Neighboring Stops is a heuristic cleanup helper and should be reviewed carefully on subtle gradients.
9. Large images and dense node trees may make sampling and preview operations feel slower.

Frequently Asked Questions (FAQ)

Question	Answer
Where do I find the add-on after installation?	Open Node Editor > Sidebar > DRH-Ramp Studio.
What Blender versions are supported?	Validated target is Blender 4.2+. Blender 5.x behavior may depend on add-on metadata and Blender API changes.
Does the add-on require internet access?	No. Normal operation is local and no external service is required.
Can I restore a converted ramp back to a native ramp?	Yes. The add-on includes restore tools for supported converted workflows.
Can I use it in Shader Editor, Geometry Nodes, and Compositor?	Yes. It support for those workflows, subject to normal context rules and some version-sensitive helper-node behavior.
Can I generate a ramp from an image?	Yes. Set Palette Source to Image, choose a readable image file, and run Generate Ramp.

Question	Answer
What happens when I convert a Cardinal or B-Spline ramp?	The add-on may use a baked sample fallback and warns before the round-trip. You can restore the original stored spline or keep the baked edits as a linear ramp, depending on the workflow.
Why does the sampler fail on Linux Wayland?	The bundled screen-capture backend does not support pure Wayland sessions. Use X11 or XWayland.
Why is screen sampling unavailable on my system?	The sampler depends on OS permissions, supported display-session backends, and being started from the Nodes N-panel.
What file types are needed for image-based ramp generation?	Use a readable image file that Blender can load successfully.
Does the add-on export reports or files?	No dedicated report or export workflow is documented. Its primary outputs are node data and temporary working assets inside Blender.
Can I rename the panel tab?	Yes. Use the N Panel Category preference in the add-on preferences.

Support, Contact, and Updates

Item	Current Status
Support Page	GitHub Support Page
Support Scope	Use the support repository for documentation, issue tracking, release notes, and general questions. Platform-specific sampler limits and documented workflow boundaries remain part of normal support context.
Issue Reporting	Report reproducible problems through GitHub Support Page /issues and include enough detail to recreate the workflow.
Release Notes / Updates	See the support repository and bundled changelog for release and packaging notes.
Discussions / Questions	Use GitHub Support Page /discussions for general workflow questions or clarification requests.
Marketplace Availability	Availability may vary across official release channels and storefronts.

Support Workflow

Use the GitHub support repository as the central public support hub. A strong support request includes the exact workflow, the active editor context, the add-on version, Blender version, operating system, and a concise description of what you expected versus what occurred.

Pre-Support Checklist

Before requesting support, please follow these requirements:

1. Add-on version and Blender version.
2. Operating system and active node editor.
3. Whether the issue affects native ramps, converted ramps, image workflows, Color Flow tools, or Color Sampler.

4. Exact reproduction steps plus expected and actual results.
5. A safe-to-share screenshot or small .blend example when possible.

Availability

This product may be available through multiple official marketplaces and storefronts, depending on release status, channel approval, and distribution strategy.

Current official storefronts:

- [Blendkit](#)

For support, documentation, issue tracking, and release updates, use the [GitHub support repository](#) as the central public support hub.

Legal, Disclaimer, and Credits

Area	Text
Credits	Product author and release maintainer: Paco Salas DRH.
Copyright Holder	Copyright (C) 2026 Paco Salas DRH.
License Notice	Distributed under the GNU General Public License, version 3 or any later version, as stated in the package notices.
Warranty Notice	Unless separately stated in writing, the software is provided without warranty.
Trademark Acknowledgement	Blender is referenced as the host application for compatibility and workflow guidance.
Third-Party Dependencies	The package redistributes python-mss / mss for Color Sampler screen capture under the MIT License; see THIRD_PARTY_NOTICES.txt.
Distribution Notice	Commercial distribution is allowed under the package notice, provided GPL terms, source availability, copyright notices, and third-party notices remain intact.

Appendix A – Glossary

Term	Definition in This Manual
Active native Color Ramp	A standard Blender Color Ramp node selected as the current target.
Converted group	A group-based ramp created by the add-on from a native Color Ramp.
Restore	The process of returning a converted group to a native ramp result.
Studio clipboard	The add-on's internal scene-level clipboard used by Copy Ramp and Paste Ramp.
Preset browser	The searchable, optionally thumbnail-based interface for curated preset palettes.
Color Sampler	The screen-stroke sampling tool that captures colors into the active ramp.
Transfer	Copying ramp data from one supported target and pasting it into another.

Term	Definition in This Manual
Interpolation	The rule used to transition between ramp colors.
Reference image	A temporary image loaded into Blender for visual guidance during palette or gradient work.
Helper nodes	Optional nodes created alongside converted groups to expose control more conveniently.
Context menu helper	A supported add-on action exposed through Blender node context menus.
Baked spline fallback	A sampled representation used when certain spline-based ramp behaviors cannot be preserved natively in the converted workflow.

Appendix B – Reference Notes

Supported working areas include Shader Editor, Geometry Nodes, and Compositor. The preset library contains 100 curated palettes. Image-based workflows use the package's documented file filter: PNG, JPG, JPEG, JP2, TIF, TIFF, BMP, TGA, EXR, HDR, WEBP, and GIF. Temporary preset preview thumbnails are written lazily to the system temp directory, and the Color Sampler uses the bundled mss dependency locally inside Blender.

Appendix C – Generator Sources and Technical Reference

This appendix summarizes generator, conversion, and sampler reference points that may help advanced users or support teams diagnose workflow differences without exposing implementation detail.

Topic	Note
Generator sources	Preset, Complementary, Analogous, Greyscale, Image, Random, and Stripes are documented package sources for Generate Ramp.
Distribution modes	The add-on documents multiple spacing rules including Linear, Ease, Smoother, Sigmoid, Mid Band, Contrast, Clustered, Center Dense, Edge Dense, Left Bias, Right Bias, Balanced, Expo In, Expo Out, Pinched, and Shoulder.
Helper node types	Generated helper node types vary by node-tree context and may use RGB or Mix-type helper nodes appropriate to Shader, Geometry Nodes, or Compositor workflows.
Linux capture backend	The add-on preferences expose Auto, XGetImage, and Xlib backend choices for Linux sampler behavior.
Image workflow dependency	Image-based palette extraction depends on readable image files and normal Blender image-loading support.
Automation / CLI / API	No dedicated command-line or public API layer is described in the packaged end-user materials.
Bundled dependency notice	Third-party attribution for the vendored mss package is included with the release archive.
Temporary files	Preset preview thumbnails can be written to the system temporary directory for UI purposes.
Reference image preview behavior	Loading a reference image does not split the workspace automatically. Use Open Image Editor Window when you want a dedicated preview area.